

Let $x \in (C \cap \bar{A})$. By Defn. 2, $x \in C \cap x \in \bar{A}$. By Defn. of AND, $x \in C$ and $x \in \bar{A}$.
 By Defn. of negation, $\neg(x \in C)$ is False. By Defn. of not belongs to, $x \notin C$ is False.
 By Defn. 5, $x \in \bar{C}$ is False. By Defn. of AND, $x \in B \cap x \in \bar{C}$ is False. By Defn. 2,
 $x \in (B \cap \bar{C})$ is False. By Defn. 3 from Sec-2.1, $x \in (B \cap \bar{C}) \cup (C \cap \bar{B})$.
 By Defn. 1, $x \in (B \cap \bar{C}) \vee x \in (C \cap \bar{B})$. By Defn. of OR, $x \in (C \cap \bar{B})$. By Defn. 2,
 $x \in C \cap x \in \bar{B}$. By Defn. of AND, $x \in C$ and $x \in \bar{B}$. By Defn. 3 from Sec-2.1,

$$\boxed{\bar{A} \subseteq \bar{B}} \quad \square$$

Let, x be an arbitrary element of $(B \cap \bar{C}) \cup (C \cap \bar{B})$, i.e. $x \in (B \cap \bar{C}) \cup (C \cap \bar{B})$.
 By Defn. 1, $x \in (B \cap \bar{C}) \vee x \in (C \cap \bar{B})$. By Defn. of OR, at least one of the
 propositions must be True.

Let, $x \in (B \cap \bar{C})$. By Defn. 2, $x \in B \cap x \in \bar{C}$. By Defn. of AND, $x \in B$ and $x \in \bar{C}$.
 By Defn. 5, $x \notin C$. By Defn. of Not belongs to, $\neg(x \in C)$. By Defn. of Negation,
 $\neg(\neg(x \in C))$ is False. By Double-Negation Law of propositions, $x \in C$ is False.
 By Defn. of AND, $x \in C \cap x \in \bar{A}$ is False. By Defn. 2, $x \in (C \cap \bar{A})$ is False.
 By Defn. 3 from Sec-2.1, $x \in (A \cap \bar{C}) \cup (C \cap \bar{A})$. By Defn. 1, $x \in (A \cap \bar{C})$
 or $x \in (C \cap \bar{A})$. From Defn. of OR, $x \in (A \cap \bar{C})$. By Defn. 2, $x \in A \cap x \in \bar{C}$. By
 Defn. of AND, $x \in A$ and $x \in \bar{C}$. By Defn. 3 from Sec-2.1, $\boxed{B \subseteq A}$

Let, $x \in (C \cap \bar{B})$. By Defn. 2, $x \in C \cap x \in \bar{B}$. By Defn. of AND, $x \in C$ and $x \in \bar{B}$.
 By Defn. of negation, $\neg(x \in C)$ is False. By Defn. of not belongs to, $x \notin C$ is
 False. By Defn. 5, $x \in \bar{C}$ is False. By Defn. of AND, $x \in A \cap x \in \bar{C}$ is False.
 By Defn. 2, $x \in (A \cap \bar{C})$ is False.

By Defn. 3 from Sec-2.1, $x \in (A \cap \bar{C}) \cup (C \cap \bar{A})$ is True. By Defn. 1,
 $x \in (A \cap \bar{C}) \vee x \in (C \cap \bar{A})$. By Defn. of OR, $x \in (C \cap \bar{A})$. By Defn. 2, $x \in C \cap x \in \bar{A}$.
 By Defn. of AND, $x \in C$ and $x \in \bar{A}$. By Defn. 3 from Sec-2.1, $\boxed{\bar{B} \subseteq \bar{A}}$

$$\therefore A = B \text{ and } \bar{A} = \bar{B} \quad \square$$